1. A glass of hot water over time loses heat energy to the surrounding and becomes cold.
2. When you mix hydrochloric acid with sodium hydroxide, it forms sodium chloride and water with evolution of heat.

In both these processes, the total energy is conserved and are consistent with the first law of thermodynamics. However, the reverse process i.e. cold water becoming hot water by absorbing heat from surrounding on its own does not occur spontaneously even though the energy change involved in this process is also consistent with the first law. However, if the heat energy is supplied to cold water, then it will become hot. i.e. the change that does not occur spontaneously and an be driven by supplying energy.

Similarly, a solution of sodium chloride does not absorb heat energy on its own, to form hydrochloric acid and sodium hydroxide. But, this process can not be driven even by supplying energy. From these kinds of our natural experiences, we have come to know that certain processes are spontaneous while the others are not, and some processes have a preferred direction. In order to explain the feasibility of a process, we need the second law of thermodynamics.

7.10.1 Various statements of the second law of thermodynamics

Entropy

The second law of thermodynamics introduces another state function called entropy. Entropy is a measure of the molecular disorder (randomness) of a system. But thermodynamic definition of entropy is concerned with the change in entropy that occurs as a result of a process.

It is defined as, $dS = dq_{rev} / T$

Entropy statement:

The second law of thermodynamics can be expressed in terms of entropy. i.e "the entropy of an isolated system increases during a spontaneous process".

For an irreversible process such as spontaneous expansion of a gas,

$\Delta S_{total} > 0$

$\Delta S_{total} > \Delta S_{system} + \Delta S_{surrounding}$

i.e. $\Delta S_{total} > \Delta S_{system} + \Delta S_{surrounding}$

For a reversible process such as melting of ice,

$\Delta S_{system} = - \Delta S_{surrounding}$

$\Delta S_{total} = 0$

Kelvin-Planck statement:

It is impossible to construct a machine that absorbs heat from a hot source and converts it completely into work by a cyclic process without transferring a part of heat to a cold sink. The second law of thermodynamics explains why even an ideal, frictionless engine cannot convert 100% of its input heat into work. Carnot on his analysis of heat engines, found that the maximum efficiency of a heat engine which operates reversibly, depends only on the two temperatures between which it is operated.

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